

TABLE I
SEMICONDUCTOR COMPONENTS USED IN MEDICAL ELECTRONICS

Product/segment	Widely-used semiconductor components
Medical imaging	Microcontrollers, microprocessors, amplifiers, data converters, digital signal processors, application-specific standard products and application-specific integrated circuits
Patient monitoring systems	Microcontrollers, sensors, ZigBee modules, application-specific integrated circuits, rectifiers, amplifiers, controllers, EEPROMs
Digital hearing aids	Microcontrollers, EEPROMs, linear accelerometers, comparators, amplifiers, audio processor integrated circuits and other application-specific integrated circuits
Infusion pumps	Microcontrollers, regulators, transistors, rectifiers, comparators, amplifiers, sensors

Source: IESA

Emerging technologies in the Indian medical electronics market

- Miniaturisation through efficient system-on-chip designs
- Wireless integrated circuits
- Efficient power management
- Intelligent sensor
- Connected information technology infrastructure

(Source: Frost & Sullivan)

of the imaging market. This market is primarily catered by import. Requirement of high precision level is the major hindrance for domestic manufacturing of imaging equipment. CT scans and MRIs are high-value, low-volume equipment used in tertiary care and high-end diagnostic labs. X-ray machines are the largest-selling imaging equipment in terms of volume.

The market for patient-monitoring systems is a mature one, which is largely driven by replacement sales. Bedside monitors account for more than 50 per cent of the patient-monitoring systems market in India. These are available with various features depending on the number of parameters displayed. Moreover, electrocardiogram machines are another important product under this category.

The patient monitoring system market is catered by domestic pro-

duction and imports. High-end patient monitoring systems are largely imported but relatively less complex products with basic features are manufactured locally.

Increasing cases of chronic respiratory diseases and kidney failure drive sales of ventilators and dialysis equipment, respectively. Ventilators and dialysis equipment together account for around 50 per cent of the therapeutics market. Ventilators are mostly used in tertiary- and secondary-level healthcare. Currently, dialysis equipment is limited to metros and tier 1 cities catering to only about five per cent of the patients in need of dialysis.

More affordable products may help the industry capture the untapped market. Therapeutics market has an almost equal contribution from domestic manufacturing and imports. High-end equipment like dialysis machines, biphasic defibrillation machines and the like are mainly imported. Lower-range therapeutic equipment are manufactured domestically.

Digital thermometers, glucometers and digital hearing aids are some fast-moving products in the handheld/homecare equipment market. Glucometers contribute to around 57 per cent of this market. An increasing number of diabetes cases drive the blood glucose monitoring equipment market.

Handheld/homecare equipment market is highly import-driven, mostly because these products are sold over the counter.

Therefore the industry focus is more on marketing rather than manufacturing. However, digital thermometers and hearing aids are being manufactured domestically by unorganised players on a very small scale.

Sourcing of semiconductor components (Table I) used in medical electronics is completely import-dependent. Cost of electronic components in medical devices varies typically between 25 and 50 per cent (Table II).

Market trends

In India, medical devices need to be made for Indian operating conditions. According to a Frost and Sullivan study, for providing healthcare, which is universal, affordable and preventive in nature, the Indian medical electronics industry is shifting towards developing miniaturised, multifunctional, ultra-low-power, portable and reduced-cost devices that could be used for patient monitoring, imaging, implantable, therapeutic and surgical requirements.

The wearable devices category, which is still a developing field in India, can potentially aid physicians to offer extended care outside the clinical environment. This would bring higher efficiency and increased access across tier 2 and tier 3 cities and rural India, while providing improved healthcare delivery to urban India.

Policy support for Make in India

With the current Make in India vision in the country, there is a huge opportunity for growth and expansion of the medical electronics industry. Union Finance Minister, Arun Jaitley, in his budget speech, urged the industry to invest in manufacturing medical equipment in India. Prime Minister Narendra Modi also made a statement that medical equipment was gaining increasing importance in the field of